Missing Value Detection Report

Project Information

Item	Value
Project Name	EnergyPulse - Smart Energy Consumption Analytics
Module	AID-3 Scalable Chunk Pipeline & Preprocessing
Deliverable	Missing Value Detection Report
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Version	1.0

1. Objective

Identify and quantify missing values within the UCI Household Electric Power Consumption dataset before preprocessing and imputation.

2. Dataset Scope

Dataset contains 2,075,259 records, 9 features, analyzed using chunk-based processing with chunk size 100,000.

3. Initial Sample Assessment

Missing values were detected in two forms: question mark (?) symbols and standard NaN values.

4. Missing Value Analysis Results

25,979 missing records were detected across six primary energy measurement columns.

5. Missing Rows Analysis

Rows containing missing values: 25,979 (1.2518% of dataset).

6. Data Quality Assessment

More than 98% of records are complete. Missing values are concentrated in the same records.

7. Engineering Recommendations

Replace ? with NaN, convert numeric columns, preserve records, and perform Seasonal Median Imputation.

8. Conclusion

Missing values affect 25,979 records and should be handled through imputation rather than deletion.

Missing Value Statistics

Column	Missing Count	Missing %
Global_active_power	25,979	1.2518%
Global_reactive_power	25,979	1.2518%
Voltage	25,979	1.2518%
Global_intensity	25,979	1.2518%
Sub_metering_1	25,979	1.2518%
Sub_metering_2	25,979	1.2518%